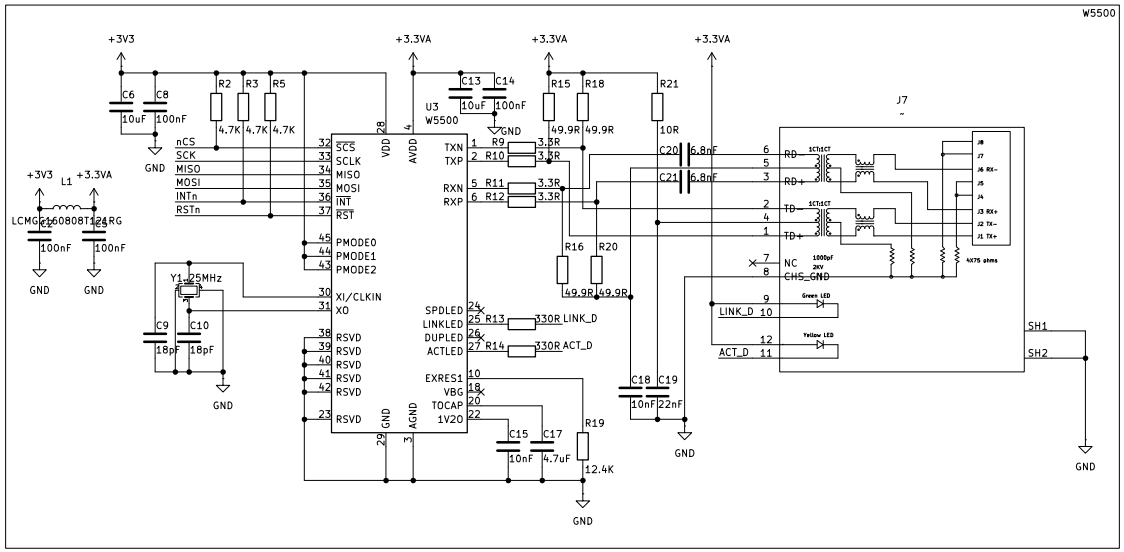
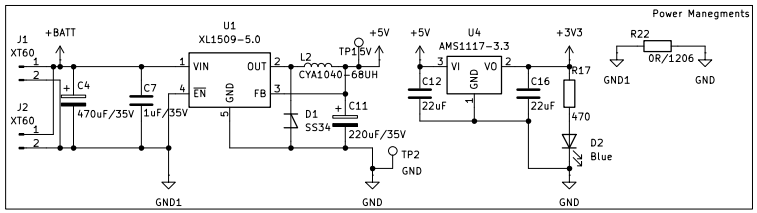
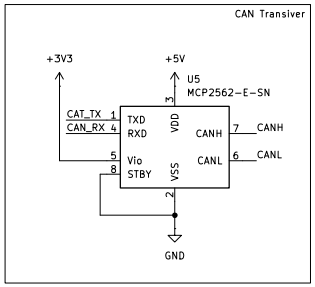
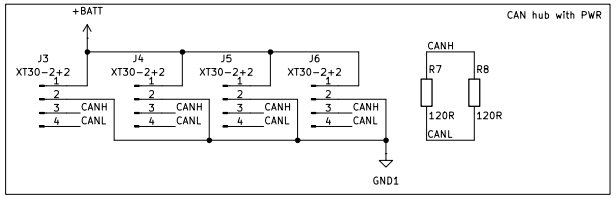
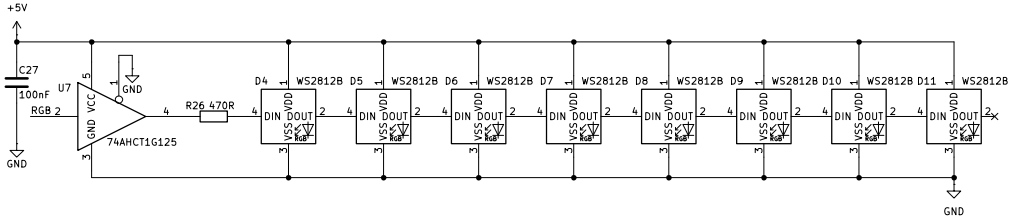
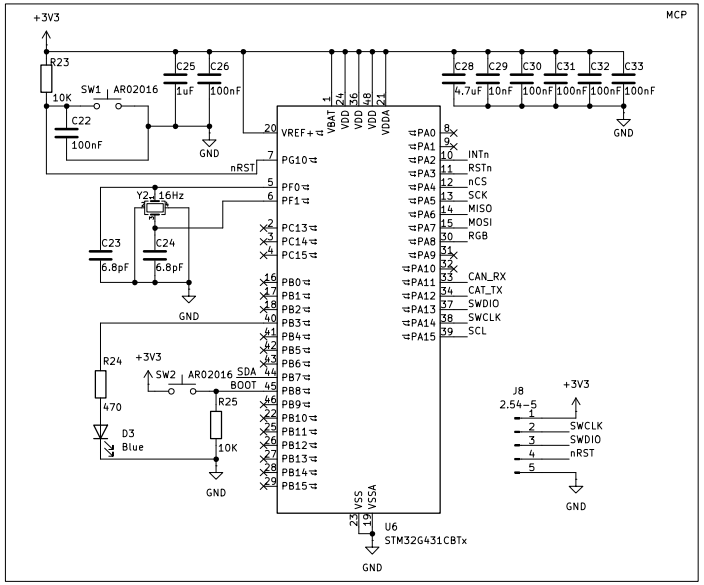

W5500
+3V3
+3.3VA
C6 10uF
C8 100nF
R2 4.7K
R3 4.7K
R5 4.7K
C13 10uF
C14 100nF
R15 49.9R
R18 49.9R
R21 10R
U3
W5500
nCS
SCK
MISO
MOSI
INTn
RSTn
SCS
SCLK
MISO
MOSI
INT
RST
VDD
AVDD
GND
TXN
TXP
RXN
RXP
R9 3.3R
R10 3.3R
R11 3.3R
R12 3.3R
C20 6.8nF
C21 6.8nF
RD
RD+
TD-
TD+
NC
CHS_GND
J7
L1
LCMG2160808T-2RG
C5 100nF
C7 100nF
Y1 25MHz
C9 18pF
C10 18pF
PMODE0
PMODE1
PMODE2
XI/CLKIN
XO
RSVD
R16 49.9R
R20 49.9R
SPDLED
LINKLED
DUPLED
ACTLED
R13 330R LINK_D
R14 330R ACT_D
Green LED
Yellow LED
LINK_D
ACT_D
EXRES1
VBG
TOCAP
1V2O
C18 10nF
C19 22nF
C15 10nF
C17 4.7uF
R19 12.4K
SH1
SH2

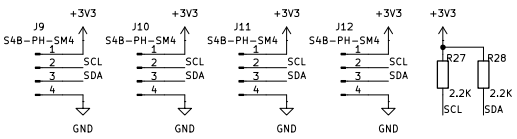

Power Manegments
J1
XT60
J2
XT60
+BATT
U1
XL1509-5.0
VIN
EN
GND
OUT
FB
L2
CYA1040-68UH
TP1
+5V
C4 470uF/35V
C7 1uF/35V
D1
SS34
C11 220uF/35V
TP2
GND1
U4
AMS1117-3.3
VI
VO
C12 22uF
C16 22uF
+3V3
R17 470
D2
Blue
R22 0R/1206
GND1


CAN Transiver
+3V3
+5V
U5
MCP2562-E-SN
CAT_TX
CAN_RX
TXD
RXD
Vio
STBY
VDD
VSS
CANH
CANL
GND


CAN hub with PWR
+BATT
J3
XT30-2+2
J4
XT30-2+2
J5
XT30-2+2
J6
XT30-2+2
CANH
CANL
R7 120R
R8 120R
GND1


+5V
C27 100nF
U7
74AHCT1G125
RGB
R26 470R
D4 WS2812B
D5 WS2812B
D6 WS2812B
D7 WS2812B
D8 WS2812B
D9 WS2812B
D10 WS2812B
D11 WS2812B
DIN
DOUT
VDD
VSS
GND


MCP
+3V3
R23 10K
SW1 AR02016
C22 100nF
C25 1uF
C26 100nF
C28 4.7uF
C29 10nF
C30 100nF
C31 100nF
C32 100nF
C33 100nF
VBAT
VDD
VDDA
VREF+
nRST
PG10
PF0
PF1
PC13
PC14
PC15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
INTn
RSTn
nCS
SCK
MISO
MOSI
RGB
CAN_RX
CAT_TX
SWDIO
SWCLK
SCL
SDA
Y2 16Hz
C23 6.8pF
C24 6.8pF
R24 470
D3
Blue
SW2 AR02016
BOOT
R25 10K
J8
2.54-5
nRST
VSS
VSSA
U6
STM32G431CBTx


J9
S4B-PH-SM4
J10
S4B-PH-SM4
J11
S4B-PH-SM4
J12
S4B-PH-SM4
+3V3
SCL
SDA
GND
R27 2.2K
R28 2.2K